

AI Study Assistant – Project

Project files converted into a readable PDF document.

File: study -assistant.py

```
from tools.calculator import add, subtract, multiply, divide
from pathlib import Path

def get_student_memory():
    memory_file = Path("memory/student_memory.txt")

    if memory_file.exists():
        return memory_file.read_text()

    return "No student memory found."

def update_student_memory(topic):
    memory_file = Path("memory/student_memory.txt")

    memory_file.write_text(
        f"Student is learning Python.\n"
        f"Current level: Beginner.\n"
        f"Current topic: {topic}.\n"
    )

    print("Memory updated successfully!")

def load_notes():
    with open("data/python_notes.txt", "r") as file:
        return file.read()

def search_notes(question, notes):
    stop_words = {"what", "is", "a", "an", "the", "of", "in", "to", "and"}

    question_words = {
        word for word in question.lower().split()
        if word not in stop_words
    }

    sentences = notes.replace("\n", " ").split(".")
    scored_results = []

    for sentence in sentences:
        sentence = sentence.strip()

        if not sentence:
            continue

        sentence_words = set(sentence.lower().split())
        score = len(question_words & sentence_words)

        if score > 0:
            scored_results.append((score, sentence))

    scored_results.sort(reverse=True)

    return [sentence for score, sentence in scored_results]

def create_quiz(topic):
    print(f"\nQuiz on {topic}")
    print("-----")

    answer1 = input("Q1. What is the basic concept of this topic? ")

    if answer1.strip():
        print("Answer recorded!")
    else:
        print("Please enter an answer.")

    answer2 = input("Q2. Give one example: ")
    if answer2.strip():
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        print("Answer recorded!")
    else:
        print("Please enter an answer.")

    answer3 = input("Q3. Why is this topic important? ")

    if answer3.strip():
        print("Answer recorded!")
    else:
        print("Please enter an answer.")

    print("\nQuiz completed!")

print("=====")
print("      AI LEARNING & STUDY ASSISTANT")
print("=====")

notes = load_notes()
student_memory = get_student_memory()

print("\nStudent Memory:")
print(student_memory)

current_topic = "Python"
while True:

    print("\n1. Ask a question")
    print("2. Show course material")
    print("3. Generate a quiz")
    print("4. Update student memory")
    print("5. Calculator")
    print("6. Exit")

    choice = input("\nEnter your choice: ").strip()

    if choice == "1":

        question = input("Enter your question: ")
        results = search_notes(question, notes)

        print("\nAnswer from course material:")

        if results:
            for result in results:
                print("-", result)
        else:
            print("Sorry, answer not found in course material.")

    elif choice == "2":

        print("\nCourse Material:")
        print("-----")
        print(notes)

    elif choice == "3":

        topic = input("Enter the quiz topic: ")
        create_quiz(topic)

    elif choice == "4":

        topic = input("Enter current topic: ")
        update_student_memory(topic)

    elif choice == "5":

        a = float(input("Enter first number: "))
        b = float(input("Enter second number: "))

        print("\nChoose operation:")
        print("1. Add")
        print("2. Subtract")
        print("3. Multiply")
        print("4. Divide")

        operation = input("Enter operation: ")

        if operation == "1":
            print("Answer:", add(a, b))

        elif operation == "2":
            print("Answer:", subtract(a, b))
        elif operation == "3":

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        print("Answer:", multiply(a, b))

    elif operation == "4":
        print("Answer:", divide(a, b))

    else:
        print("Invalid operation.")

elif choice == "6":

    print("\nThank you for using AI Study Assistant!")
    break

else:

    print("\nInvalid choice.")
```

File: tools/calculator.py

```
def add(a, b):  
    return a + b  
  
def subtract(a, b):  
    return a - b  
  
def multiply(a, b):  
    return a * b  
  
def divide(a, b):  
    if b == 0:  
        return "Cannot divide by zero"  
    return a / b
```

File: memory/student_memory.txt

Student is learning Python.

Current level: Beginner.

Current topic: functions.

File: data/python_notes.txt

Python is a high-level programming language.

Python is easy to learn and widely used in web development, data science, automation, artificial intelligence and machine learning.

Variables are used to store data.

A function is a reusable block of code.

A list is used to store multiple values.

Python uses indentation to define blocks of code.

File: data/ai-study assistant project/python_notes.txt

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